Low-Cost High-Speed Air-to-Air Missile RFI
Agency: Department of the Air Force
Office: Air Force Materiel Command
Location: AFLCMC/EB – Eglin Air Force Base

Notice Type: Request for Information
NAICS Code: 336414 – Guided Missile and Space Vehicle Manufacturing

DISCLAIMER:

THIS IS A REQUEST FOR INFORMATION (RFI) ONLY. This RFI is issued solely for information and planning purposes – it does not constitute a Request for Proposal (RFP) or a promise to issue an RFP in the future. This request for information does not commit the Government to contract for any supply or service whatsoever. Further, the Government is not seeking proposals at this time and will not accept unsolicited proposals. Vendors are advised that the Government will not pay for any information or administrative costs incurred in response to this RFI; all costs associated with responding to this RFI will be solely at the interested party's expense. Not responding to this RFI does not preclude participation in any future RFP that may be issued. This notice does not in any way restrict the Government to an ultimate acquisition approach. The Government encourages all responsible businesses, including small businesses, to respond to this RFI. Respondents to this RFI may be requested to provide additional information/details based on their initial submittals. Any submittals will not be returned to the sender.

SUBJECT:

The Armament Directorate is exploring industry potential for delivering a Low-Cost High-Speed Air-to-Air Missile (referred to herein as "the missile") to support Air Force needs around the globe. The purpose of this Request for Information (RFI) is to identify industry partners with ability and capacity to deliver a prototype with low Integration and Non-Recurring Engineering (NRE) costs achieved by eliminating or limiting use of critical technology elements (CTE) and making use of High Technical Readiness Level (High-TRL) and High Manufacturing Readiness Level (High-MRL) weapon assemblies, subassemblies, and components in accordance with weapons requirements as listed below. The desire is to maximize reuse of existing components.

DESCRIPTION:

Industry is encouraged to provide responses for the missile with a form fit factor similar to the AIM-120 AMRAAM to provide a low-cost solution with maximum range. Industry is also encouraged to provide low-cost solutions for the missile at half the size of an AMRAAM to double aircraft loadout while maximizing range in the design.

INTEREST:

Successful responders must be capable of obtaining any necessary licenses or data rights from the original manufacturer, providing a Rough Order of Magnitude (ROM) cost, and outlining an associated delivery schedule or lead times for available capabilities.

For questions, contact Lt Col Clair (YETI) Corbin (Program Manger Lead) at clair.corbin@us.af.mil and Jill Harms (Contracting Lead) at jill.harms.1@us.af.mil.

Questions are due by 16 May 2025, by 1700 CST. All questions will be answered and posted publicly as an amendment to this RFI.

An ***Ask me Anything*** will be hosted virtually on 14 May 2025 @ 1300CST. The invite will be posted as an amendment to this posting and available to all interested parties. All questions and answers from the *Ask me Anything* will be posted as an amendment to this posting.

Responses are due by Monday 2 June 2025, by 1700 CST to Lt Col Clair (YETI) Corbin and Jill Harms.

Statement of Capabilities (SOC):

Interested and capable sources are encouraged to respond to this RFI with a Statement of Capabilities that provides specific relevant knowledge and experience in the following areas as it pertains to satisfying the requirements detailed below in section a-e. Interested and capable parties are encouraged to consider teaming solutions as the USG finds value in a best of breed approach for this capability. Sections a-d below illustrate the capability as the USG understands the requirement and responses should address each of the sections, however the USG is interested in any solution that offers value to the USG even if it does not address all of the factors. Failure to meet these specifications in their entirety does not preclude an interested party from submitting a response that meets the overall intent of this RFI.

The Statement of Capabilities should address these factors:

a. Targets: Air-to-Air (Not intended as priority order: 4th Gen Aircraft with moderate maneuvers; large aircraft such as AEW, bombers, and refuelers)
b. Aircraft Integration Targeted at: F-16, F-15E, Collaborative Combat Aircraft (CCA). Also state if other launch platforms are being considered, e.g., Longshot, ground launch, etc.
c. Production Capacity: 1,000 AURs NLT 24 months from contractor award
d. Target Cost for All Up Round (AUR): <= $250K (1,000 AURs). Vendors may provide a path to achieve this AUR cost from the prototyping phase to production. If vendors have price estimates for their high-TRL, high-MRL, non-developmental, low integration NRE prototype and production estimates that may not meet this price point, vendors are encouraged to submit their estimated prices for AUR designs and prototypes through production that are above this price point.

Tradespace:

- Speed: Supersonic to hypersonic speed regimes
- Time of flight (from launch to target intercept)
- Target Sets
- Range
- Probability of Kill (Pk)
- Reliability
- Data Links

Assumptions:

- Target Handoff. High-confidence target handoff from aircraft.
- Maturity of Hardware. High-TRL, High-MRL seekers which assume high-confidence handoff from aircraft. Low-cost Integration NRE, no critical technology elements.
- Aircraft Integration. Universal Armament Interface (UAI) or Common Flexible Weapons Interface (CFWI) for aircraft to weapon interface. Note: CFWI contains the UAI interface, an Anti-Tamper interface, and an air-to-air datalink (if needed for proposed design).
- Open System Architecture. Weapon design in compliance or a clear path to achieve compliance with the Weapon Open System Architecture (WOSA) Reference Architecture.
- Weapons Government Reference Architecture. Weapon design to be based in Model Based Systems Engineering (MBSE) principles with use of Weapons GRA and derivation of the Implementation Model from Weapons GRA requirements. Alternatively, vendors may present their plans for a clear path to develop an Implementation Model derived from Weapon GRA requirements.
- Modeling & Simulation. Six degree of freedom (6DOF) simulation which accurately models performance of weapon design. Planned full verification and validation of the missile 6DOF.

Test:

Interested parties are encouraged to consider alternate solutions for testing with the end state integration platform being the F-16, F-15E, or Collaborative Combat Aircraft (CCA). If considering other test platforms, please state.

Schedule:

For planning purposes, the USG could foresee a schedule that would include two (2) six-month design cycles, culminating in a 12-month follow-on for production. Interested businesses are encouraged to use this information to understand the potential pace of the requirement.

Submission Details:

Interested businesses should submit a brief capabilities statement package (no more than ten (10) 8.5 X 11.5-inch pages, font no smaller than 10 point) demonstrating the ability to perform the required capabilities and meet pricing. All responses shall include Company Name, Company Address, Company Business Size, and Points-of-Contact (POC) including name, phone number, and email address, and Company CAGE code.

The Statement of Capabilities should illustrate the ability to meet the requirements a-e, identify trade space decisions made (highlighting any high MRL/TRL components), provide the estimated All Up Round cost based on a production lot of 1,000 rounds, and provide details describing the vendor's ability to deliver 1,000 rounds within 24 months after contract award. Additionally, the USG is requesting a ROM that includes the vendor's plan to deliver the prototype as well as the AUR/production costs. If the vendor anticipates using a teaming approach that should be identified in the submission with a clear breakout of work to be completed by each vendor. Additionally, if a vendor is taking an alternative approach that does not directly align with the SOC sections a-e but aligns with the intent of this RFI, the vendor should clearly outline how the vendor's solution will provide a capability to the USG and address all other sections of this paragraph, with special attention on tradespace decisions made.